Class 6

Two-player Zero-sum games

A game $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ is called a two-player zero-sum game if

- $N = \{1, 2\}$
- $u_1(s_1, s_2) + u_2(s_1, s_2) = 0 \quad \forall (s_1, s_2) \in S_1 \times S_2$

U , NU

$u(U, R) = 3 \qquad u(NU, R) = -2$

$u(U, NR) = 1 \qquad u(NU, NR) = 3$

DM \ N	R	NR
U	3	1
NU	-2	3

$$\underline{v_1} = \max_{s_1 \in S_1} \min_{s_2 \in S_2} u(s_1, s_2) \qquad \Big| \; u = u_1$$

$$\underline{v_2} = \max_{s_2 \in S_2} \min_{s_1 \in S_1} \left(-u(s_1, s_2)\right)$$

$$= \max_{s_2 \in S_2} \left[- \max_{s_1 \in S_1} u(s_1, s_2) \right)$$

$$z = -\min_{s_2 \in S_2} \max_{s_1 \in S_1} u(s_1, s_2)$$

$$\underline{v} = \max_{s_1 \in S_1} \min_{s_2 \in S_2} u(s_1, s_2) \qquad \bar{v} = \min_{s_2 \in S_2} \max_{s_1 \in S_1} u(s_1, s_2)$$

In general, $\underline{v} \leq \bar{v}$

- $\underline{v}$: min amount P-1 can guaranteed to get
- $\bar{v}$: max amount P-2 can guarantee to pay

Player-1: the max min strategy or any strategy that ensures him a utility of $\underline{v}$

s_1 is max min strategy $\iff s_1 \in \arg\max_{s_1 \in S_1} \left[\min_{s_2 \in S_2} u(s_1, s_2) \right]$

Player-2 the max min strategy is any strategy that ensures a maximum of $\bar{v}$ as his payment.

In this context, for player 2, we call the max-min strategy as min-max strategies.

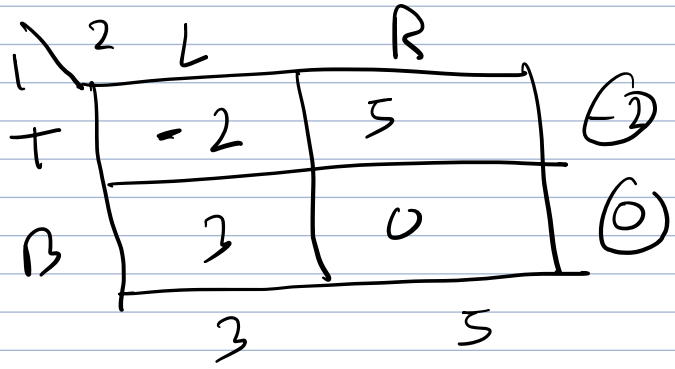
1 \ 2	L	C	R	
T	3	-5	-2	(-5)
M	1	4	1	(1)
B	6	-3	-5	(-5)
	6	4	1	

$\underline{v} = 1$

$\bar{v} = 1$

max-min strategy M

min-max strategy R



$\underline{v} = 0$

$\bar{v} = 3$

Defn For a two-player zero-sum game if $\underline{v} = \bar{v} = v$ then we say the game has a value v. And, the max-min and min-max strategies are called optimal strategies.

$$\underline{v} \leq \bar{v}$$

$u(s_1^*, s_2^*)$ where s_1^* is a max-min s. s_2^* is a min-max s.

$$\underline{v} \leq u(s_1^*, s_2^*) \leq \bar{v}$$

if $\underline{v} = \bar{v} = v$

$$u(s_1^*, s_2^*) = v$$

$$u(s_1, s_2^*) \leq v \qquad u(s_1^*, s_2) \geq v$$

Nash Equilibrium A strategy profile $(s_1^*, s_2^*, \ldots, s_n^*)$ is called a NE if $\forall i$

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i \in S_i$$

For a two-player zero-sum game, if value exists.

then (s_1^a, s_2^b) is an NE where s_0^a is a max-min strategy of Player 1 and s_2^b is a min-max strategy for Player 2.

M \ W	F	D
F	(2,1)	(0,0)
D	(0,0)	(1,2)